INCH–POUND

MIL–DTL–52011G(EA)
w/AMENDMENT 1
6 September 2007

SUPERSEDING
MIL–DTL–52011G(EA)
8 July 2003

DETAIL SPECIFICATION

FILTER, PARTICULATE, 12 CFM, M13

Inactive for new design after 19 April 1996

This specification is approved for use by the U.S. Army Edgewood Chemical Biological Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of particulate filter.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Edgewood Chemical Biological Center, ATTN: AMSRD–ECB–ENA–S, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010–5424 or emailed to SpecsTeam@apgea.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL−STD−282 − Filter Units, Protective Clothing, Gas−Mask Components and Related Products: Performance−Test Methods

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111−5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

U.S. ARMY EDGEWOOD CHEMICAL BIOLOGICAL CENTER

DRAWINGS

5−19−853 − Filter, Particulate, 12 CFM, M13

(Copies of these documents are available from U.S. Army Edgewood Chemical Biological Center, ATTN: AMSRD−ECB−ENA−D, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010−5424.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on 5−19−853 and on the subsidiary drawings shall conform to the specifications listed thereon, and to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the particulate filter shall conform to the specifications and drawings listed on 5−19−853 and subsidiary drawings.

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3.2 Manufacture and assembly. The particulate filter shall be manufactured and assembled as specified on Drawing 5−19−853.

3.3 Resistance to airflow. The resistance to airflow shall be no greater than 0.8 inch of water gage (iwg) at an airflow rate of 12 standard cubic feet per minute (scfm) when tested as specified in 4.4.1.

3.4 Filtration efficiency. The filters shall be no less than 99.97% efficient in the removal of aerosols with a mean diameter of 0.3 microns when tested in accordance with 4.4.2.

3.5 Air leakage. Air leakage shall be not greater than 0.91 cubic inch per minute when the filter is internally pressurized to a pressure of not less than 12 inches of water when tested as specified in 4.4.2.

3.6 Resistance to rough handling. The filters shall meet the requirements of 3.3, 3.4, and 3.5 after rough handling for 15 minutes at an amplitude of 3/4 inch with the filter face, pleats, and separators in a vertical position, when tested as specified in 4.4.4.

3.7 First article. When specified (see 6.2), a sample shall be subjected to the first article inspection in accordance with 4.2.

3.8 Workmanship. The particulate filter shall be free from chipped, burred, or bent metal sections; foreign matter, such as oily or viscous substances; and rips, tears, or holes in the filtering medium.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Sample. The first article sample shall consist of 10 particulate filters manufactured prior to the start of regular production and using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.2.2 Inspections to be performed. As determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings.

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4.2.3 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.3 Conformance inspection.

4.3.1 Lotting. A lot shall consist of the filters produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and offered for acceptance at one time; however, no more than one lot of filter material shall be represented in any one lot of finished filters. Each lot shall be identified by an alphanumeric lot number. The lot number shall include a manufacturer's identification symbol consisting of 3 alpha characters, a numeric code identifying the year of production, a code or abbreviation that signifies the month of production, and an interfix−serial number. The interfix−serial number shall change if there is a change in the design, manufacturing process, materials, suppliers, production run, or if a new contract is used.

4.3.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.3.5 and, when specified, table I or table II. Samples shall be selected at random.

TABLE I. Sampling for rough handling

Lot size	Sample size
2 to 15	2
16 to 25	3
26 to 90	5
91 to 150	8
151 to 500	13
501 to 1,200	20
1,201 to 10,000	32
10,001 to 35,000	50
35,001 to 500,000	80
Over 500,000	125
Accept the lot represented on zero nonconforming characteristics and reject the lot on one or more nonconforming characteristics for all sample sizes.	

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4.3.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. Sample filters shall be examined and tested in accordance with the classification of characteristics in 4.3.5. If a filter is found that does not conform to any characteristic inspected 100%, such as filtration efficiency and airflow resistance, the non−conforming filter shall be rejected and removed from the lot. For characteristics other than filtration efficiency and airflow resistance, failure of any sample filter to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented (see footnote, table I).

4.3.4 Critical characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

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TABLE II. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

4.3.5 Classification of characteristics. Conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table II for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.5	Filter, particulate, 12 cfm, M13		5–19–853
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical				
1	Filtration efficiency	100 percent inspection	3.4	4.4.3
2	Air leakage	100 percent inspection	3.5	4.4.2
Major				
101	Airflow resistance	100 percent	3.3	4.4.1
102	Resistance to rough handling	Table I	3.6	4.4.4
103	Component evident	Table II, level VII	3.1.2	VI
104	Component correctly assembled	Table II, level VII	3.2	VI
105	Overall dimensions correct	Table II, level VII	3.2	CE
106	Marking evident and correct	Table II, level VII	3.2	VI
107	Workmanship	Table II, level VII	3.8	VI
Minor				
201	Marking legible	Table II, level X	3.2	VI

NOTES:
CE – Commercial inspection equipment
VI – Visual inspection

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4.4 Tests.

4.4.1 Airflow resistance. The airflow resistance shall be determined at the airflow of 12 cfm. The measured pressure drop across the particulate filter, when corrected to standard conditions of 21˚C (70˚F) and 1 atm (1013 mbar), shall be no greater than that specified in 3.3. The up−stream and down−stream static pressure measuring tubes shall be as close as possible to the filter and shall not be on a section of duct that has a changing cross sectional area. Test for airflow resistance as follows:

(a) Connect the filter to a source of forced air.
(b) Set the flow of air through the filter to 12 cfm.
(c) Record the barometric pressure.
(d) Measure and record the air stream temperature.
(e) Determine and record the difference between up−stream and downstream static pressure.

If the recorded air stream temperature and barometric pressure is different than 21˚C (70˚F) and 1 atm, calculate and record the air flow resistance (DP (cal)) using the equation below to correct the measured airflow resistance to airflow resistance at standard conditions:

$$DP\ (cal) = DP\ (measured) \times P\ (test) \times 86.21 \div [(492 + 1.8T)^{1.768}]$$

where:

DP (cal) – air flow resistance corrected to standard conditions in millimeters of water gage (mm wg).
P (test) – barometric pressure at time of test in millimeters of mercury (mm Hg).
DP (measured) – air flow resistance from test measurement in millimeters of water.
T – temperature of air stream flowing through the filter in degrees C.

Note: Correction for nonstandard conditions is not required when it is clear that the passage or failure of the filter is not in doubt.

4.4.2 Air leakage. The sample filters shall be tested for leakage using a test apparatus which shall have two flat (within .005 inch) smooth (32 micro inch) surfaced metal plates at least .25 inch thick that can be clamped to the gaskets on the inlet and outlet of the filter. The clamping mechanism shall be controlled so that the total combined compression of the two gaskets does not exceed .120 inch. There shall be provisions in one of the plates which shall be fixed to the base of the apparatus for pressurized air to be introduced into the filter under test through a flow meter capable of measuring a flow of 0.91 ± 0.06 cubic inches per minute of air flow. There shall also be provisions for measuring the pressure inside the filter between plates through the use of a manometer or pressure gage capable of measuring pressures of 12.00 inches of water with an accuracy of ± .06 inches of water. The internal pressure measurement shall be taken through a separate passage through one of the two plates from the passage that is used to introduce the measured flow of air. The apparatus shall also have a flow regulation valve by which the air flow can be controlled. The test shall be conducted by clamping the filter to be tested between the two

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metal plates, opening the air flow control valve, and adjusting the flow until a pressure of at least 12 inches of water is established within the filter. The flow rate that is required to maintain this pressure is the air leakage value. The filter will be determined to be defective if the air leakage value exceeds 0.91 cubic inches per minute.

4.4.3 Filtration efficiency. The test apparatus used to measure the filtration efficiency shall be capable of maintaining a stable concentration of aerosol with a mean particle size of 0.3 microns with a steady flow through the filter of 12 scfm. The test apparatus shall have the capability to determine the concentration of aerosol both upstream and down stream of the filter within 0.005%. The downstream sample point shall be located such that changing of its location across the ductwork does not cause a significant change in the measured concentrations. ~~This shall be verified using a filter with a known measurable leak.~~ The filter shall be placed in the test apparatus such that the flow of air containing the challenge aerosol is in the direction indicated by the marking on the filter, and the flow of aerosolized air shall be maintained at 12 ± 3 scfm. This flow shall be maintained for not less than 1 minute or more than 2 minutes. The filter shall be rejected if the downstream concentration exceeds 0.03 percent of the upstream concentration which indicates that the filtration efficiency is less than 99.97 percent. (See note 6.5)

4.4.4 Resistance to rough handling. The filters shall be rough handled in accordance with Method T105.10 of MIL−STD−282, and then tested for resistance to airflow, air leakage, and filtration efficiency in accordance with 4.4.1, 4.4.2, and 4.4.3 respectively.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When ~~actual~~ packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain ~~requisite~~ packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD−ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The particulate filter covered by this specification is intended to be used in an air purifier.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification

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(b) DELETED
(c) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(d) Shelf life code
(e) Level of packaging (see section 5).

6.3 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.4 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

6.5 Filtration efficiency testers and aerosol materials. The TDA 100 Filter Penetration Tester manufactured by Air Techniques Inc. has been found to be capable of performing the filtration efficiency test when operated with Poly−Alpha Olefin (common trade names are Ethyl Flow 164 and Emery 3004) or Dioctylphthalate (DOP) as the aerosol agent. DOP is not recommended because of its potential as a carcinogen. Filtration efficiency is calculated by subtracting the measured percent penetration from 100 percent.

$$\%P = [(\text{PPM downstream})/(\text{PPM upstream})] \times 100\%$$
$$\text{Eff} = 100\% - (\%P)$$

Where:
PPM − Particles Per Minute

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%P – Percent Penetration
Eff – Filtration Efficiency

Custodian:

Army – EA

Preparing activity:

Army – EA

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